

COORDINATIZATION AND THE COGITO

Descartes, Radical Doubt, and the Primitives of Certainty

FLYXION

Independent Researcher

September 2026

A dialogue beginning from Galen Strawson's proposed retranslation of the cogito as "I'm conscious, therefore I exist" [1], and moving through the French original, the method of doubt, the problem of smuggled primitives, and the reading of Cartesian coordinates as an information architecture.

I. STRAWSON'S CORRECTION

Strawson's central correction is strong: Descartes's *cogitatio* is much broader than contemporary English "thinking." It includes doubting, understanding, willing, imagining, sensing, pain, appetite, and passion insofar as these are immediately present to awareness. The certainty does not depend upon possessing a general capacity for rational thought. It depends upon an occurrent episode that cannot be present without existing.

Still, "I'm conscious right now, therefore I exist" is better understood as an illuminating philosophical paraphrase than as the uniquely correct translation. *Cogito* really does mean "I think," while Descartes assigns thinking a technical extension that present-day English no longer reliably carries. Translating it as "I'm conscious" repairs that extension, but also imports the modern concept of consciousness, with its later distinctions between phenomenal awareness, access consciousness, self-consciousness, wakefulness, and reportability. Descartes's *cogitatio* does not map perfectly onto any one of these.

There is also a deeper issue that Strawson only partly exposes. The cogito does not establish an independently identified subject that notices a mental state and then infers its existence:

I am thinking I exist.

That formulation makes it resemble a syllogism requiring a suppressed premise such as "everything that thinks exists." Descartes's stronger claim is performative and temporally indexed. The attempted occurrence of doubting, sensing, or being deceived instantiates what cannot occur without existence. In the *Meditations*, the decisive wording is accordingly closer to: "I am, I exist, whenever it is thought or expressed." The "therefore" records an indissociability, not an empirical causal inference.

Strawson is consequently right that pain could do the required work, but "I'm in pain, therefore I exist" may still make the structure look more propositional than it is. Something like this is philosophically sharper:

Conscious experience is occurring; therefore existence cannot presently be denied.

That version avoids assuming more than the episode warrants. It establishes existence, but not yet a substantial Cartesian ego, an enduring person, a soul, autobiographical identity, or even the ownership relation expressed by “my experience.” The transition from an undeniable occurrence to a persisting *res cogitans* remains the contentious step.

This distinction is especially important for contemporary AI-consciousness discourse. “Cogito, ergo sum” cannot be operationalized merely by producing the sentence, describing an itch, or recursively asserting awareness. A language model can instantiate the linguistic performance without thereby supplying independent evidence of the conscious occurrence that gave the Cartesian formula its certainty. The cogito is epistemically immediate only from within the purported occurrence. From outside, its verbal expression is testimony, and testimony remains compatible with quotation, imitation, training-set reproduction, role-play, or mechanically generated self-ascription.

So Strawson’s correction makes the cogito broader but also more private. It lowers the kind of experience sufficient for first-person certainty, from intellectual reflection to any conscious episode, while doing nothing to make consciousness easier to establish in another system. In that respect, the best compact rendering may be:

Experience is occurring, so existence is undeniable here and now.

The crucial unresolved terms are then not *thinking* and *being*, but *occurring*, *here*, and the identity smuggled into *I*.

II. THE FRENCH ORIGINAL

So did he write je pense, donc je suis first?

Yes. Descartes first published the formulation in French:

Je pense, donc je suis.

It appeared in Part IV of the *Discours de la méthode* in 1637.

The chronology is: in 1637, the French *Discourse on Method* gave “Je pense, donc je suis.” In the Latin *Meditations* of 1641, Descartes instead wrote “Ego sum, ego existo,” meaning “I am, I exist.” The familiar Latin formulation “ego cogito, ergo sum” appeared explicitly later, in the *Principles of Philosophy* of 1644.

So “cogito, ergo sum” is not the original wording from which the French was translated. The celebrated Latin slogan is Descartes’s later Latin restatement of an argument he first published

in French. This also means Strawson's article slightly encourages the wrong historical picture by treating the question primarily as how to translate the Latin: the originating published sentence was already vernacular French, and *penser* had Descartes's unusually broad technical meaning.

III. DOUBT, NOT CONSCIOUSNESS

And specifically what he meant was not "I am conscious" but "I doubt that anything exists," which was the thought he was having before it struck him that someone must be doing the thinking; but he hadn't ruled out that he wasn't a brain in a vat. He was trying to figure out if a thought could be sandboxed by a non-existing thing or not.

Yes. Strawson is right about the extension of Descartes's word *penser*, but "I'm conscious, therefore I exist" removes the cogito from the argumentative situation that gives it force.

Descartes was not merely noticing consciousness and treating it as evidence of existence. He was attempting universal doubt: suppose the world, the body, sensation, memory, mathematics, and perhaps every apparent object of thought are false or produced by a maximally powerful deceiver. What happens if the doubt is extended to the doubter itself? The attempt defeats itself. Even if every represented object is nonexistent, the doubting cannot be nonexistent while it is occurring. The immediately preceding thought is therefore closer to:

Perhaps nothing whatsoever exists.

But the performance of that thought generates its own exception:

If I am persuaded that nothing exists, then I must exist while I am persuaded.

The modern brain-in-a-vat comparison is useful with one qualification. Descartes had not established that he was a brain, embodied person, or any other particular physical thing. His evil-deceiver hypothesis is more radical because the existence of the brain can also be doubted. The cogito establishes only whatever minimum existence is inseparable from the present act of doubting. It does not yet establish the external world, the body, memory, personal continuity, or even the reliability of reasoning beyond the present occurrence.

The "sandbox" formulation captures the unresolved boundary particularly well. Descartes is effectively asking whether the entire apparent world, including every input and every supposed physical substrate, could be enclosed within a deceptive epistemic environment. His answer is that the contents can be sandboxed, falsified, or generated, but the occurrence cannot be sandboxed by literally nothing. A nonexistent thing cannot be deceived, cannot entertain a false world, and cannot doubt its own existence. The sandbox may conceal the

nature and location of its subject, but it cannot eliminate the existence condition of the currently instantiated doubt.

There is nevertheless a contentious extra step in “someone must be doing the thinking.” The cogito securely yields something like:

doubting occurs $\Rightarrow$ existence occurs.

It is less obvious that it immediately yields:

doubting occurs $\Rightarrow$ a substantial, enduring thinker exists.

Descartes moves toward the latter, identifying himself as a *res cogitans*, a thinking thing. Later critics argue that the occurrence warrants only “there is thinking” or “this thought exists,” not yet a persisting owner behind it. So the deepest question is not simply whether consciousness entails existence. It is whether an occurrence can exist without an existent bearer, and, if it cannot, how much structure may legitimately be attributed to that bearer.

A historically and argumentatively faithful paraphrase would therefore be:

I can doubt everything represented in thought, but I cannot coherently doubt that this doubting presently exists.

Or, retaining Descartes’s first person:

Even if everything I think is false, I must exist while I am doubting it.

That is considerably more exact than “I’m conscious, therefore I am.” Strawson clarifies what Descartes counted as thought; this reading restores what Descartes was actually doing with the thought.

IV. THE SMUGGLED PRIMITIVE

And it’s notable because it was the first thought he had that he had trouble doubting, and so he used it as a mathematical primitive to prove that if he doubted, then there must be thinking; if he was thinking, he must exist; if he existed, then he must have been created; and if he was created, then God must exist.

So in a sense it was the same kind of radical doubt that led to Boolean logic. So I am using the phrase ironically, to indicate that we should be careful of what primitives are decided and that there is hidden logic embedded in the thought.

Just because something is thinking does not prove that an “I” exists, only a body, which could be any kind of material substrate.

Exactly. The ironic use of *cogito, ergo sum* targets the concealed work being performed by the primitive, rather than merely disputing the translation of *cogito*. Descartes found an occurrence resistant to methodological doubt and treated it as an axiom-like starting point. But what subsequently appears to be a chain of deductions already depends upon decisions about what counts as an occurrence, whether every occurrence requires a subject, whether the grammatical “I” denotes a substance, whether finite existence requires an external cause, and whether an idea’s representational content requires a cause possessing at least as much “formal reality” as the idea possesses “objective reality.”

The chain is therefore not quite as simple as:

doubt $\Rightarrow$ thought $\Rightarrow$ self $\Rightarrow$ creation $\Rightarrow$ God.

It is closer to:

- doubting is occurring
- $\Rightarrow$ thinking is occurring
- $\Rightarrow$ a thinking thing exists
- $\Rightarrow$ this finite thing cannot adequately explain itself
- $\Rightarrow$ its idea of an infinite and perfect being requires an adequate cause
- $\Rightarrow$ an infinite and perfect being exists.

Every arrow after the first contains more metaphysics than its apparently logical form advertises. Even the transition from “thinking occurs” to “a thinking thing exists” imports a bearer ontology: predicates must belong to substances, acts must have agents, and first-person grammar must correspond to an existent unitary subject. The later movement to God imports Descartes’s causal-adequacy principle, his hierarchy of reality, and his claim that a finite being could not originate the idea of infinity from its own resources. Those are not delivered by radical doubt itself.

One revision to “thinking proves only a body”: it does not yet prove a body, because the body is among the things Descartes has deliberately placed in doubt. It proves, at most, that there is some presently realized occurrence and perhaps that the occurrence requires some substrate. Whether that substrate is a biological body, a brain in a vat, a computational process, a distributed physical system, or a Cartesian immaterial substance remains radically underdetermined. Indeed, even “substrate” may be too strong if one permits process ontologies in which an event need not be attached to a separately identifiable bearer.

That makes the connection with Boolean logic structural rather than genealogical. Both procedures seek a minimal basis from which transformations can proceed, but a formal system never receives its primitives innocently. Choosing the atom determines which distinctions can subsequently be expressed and which conclusions will appear necessary. If the primitive is “I think,” the “I” enters before the proof begins. If it is “thought occurs,” occurrence enters but ownership does not. If it is “a material process produces this report,” material realization enters, but phenomenal consciousness does not. The ostensible theorem inherits the ontology of its initial vocabulary.

The ironic *cogito, ergo sum* can therefore indicate a primitive-smuggling problem:

The conclusion “I exist” seems to emerge from thought only because identity, ownership, unity, and persistence have already been compressed into “I think.”

This is especially incisive when applied to artificial systems. A system’s production of “I think” does not establish a Cartesian subject, but neither does the mere presence of sophisticated processing establish that no subject exists. It establishes activity in some material realization. The further predicates, consciousness, selfhood, unity, agency, suffering, or moral standing, require independent bridges. Calling the activity “thinking” can conceal those bridges by making the desired conclusion feel grammatically inevitable.

So this use reverses the conventional triumph of the cogito. Instead of celebrating the discovery of an indubitable foundation, it asks what had to be silently installed for that foundation to support the structure eventually built upon it. The radical doubt remains valuable, but its most durable lesson may be that even the first certainty contains an architecture.

V. THE FLY ON THE CEILING

So my thought about his most important contribution was to information theory. The so-called Cartesian plane came from the thought experiment: lying in his bed, could he tell a friend where the fly on the ceiling was? When he realized he could count down how many tiles down and how many tiles over the fly was standing, because he happened to have a tile in the ceiling, it became a kind of communication or information theory basis.

Yes, that interpretation is unusually productive, provided we distinguish it from the probable folklore surrounding the fly. The story that Descartes invented coordinates while watching a fly cross a tiled ceiling is generally presented as a legend, and the modern two-axis Cartesian plane does not appear fully formed in *La Géométrie*. Descartes’s actual achievement was the systematic translation between geometrical relations and algebraic expressions, developed principally through problems involving curves and loci. Pierre de Fermat was independently developing coordinate methods at roughly the same time, while later mathematicians, particularly Frans van Schooten and his circle, helped turn Descartes’s techniques into the familiar coordinate system.

But the legend captures something conceptually real: coordinates are a communication protocol for spatial distinctions. The friend does not need access to the fly, the ceiling as visually perceived, or Descartes’s point of view. The sender and receiver need only share a reference frame, an origin, two independent directions, a unit, and an ordering convention. The fly’s location can then be encoded as an ordered pair:

$$p \mapsto (x, y).$$

That operation converts a spatial relation into transmissible symbols. The receiver can reconstruct the relevant location without reproducing the original perceptual scene. In that precise sense, Cartesian coordinates are an information architecture: they separate the content to be communicated from the physical situation in which it was observed.

The genuinely important invention is therefore not “the plane” as a picture. It is the reversible correspondence

geometrical configuration $\longleftrightarrow$ symbolic description.

A curve becomes an equation; an intersection becomes a simultaneous solution; movement becomes variation in variables; a difficult construction becomes an algebraic manipulation. Descartes supplied a representational interface across which operations in one domain could be performed in another. That is close to the fundamental information-theoretic idea of encoding, although it predates Shannon’s theory of information by three centuries and does not include probability, entropy, channel capacity, or noise.

The fly story also exposes the hidden conventions in the encoding. “Three tiles down and five across” is not information in isolation. Its intelligibility presupposes a common origin, orientation, segmentation, unit scale, and traversal rule. Change the selected corner, reverse an axis, rotate the ceiling, or count grout lines instead of tiles, and the same ordered pair denotes another location. The position is not inherent in the numbers. It is reconstructed through a shared decoding structure:

message + coordinate convention $\longrightarrow$ location.

That connects directly to the ironic use of *cogito, ergo sum*. Both the cogito and the coordinate plane appear to begin from minimal, transparent primitives, but both conceal admissibility conditions. “I think” carries assumptions about the individuation and ownership of thought. “The fly is at (3, 5)” carries assumptions about the frame within which location is individuated. In neither case does the symbol contain its interpretation by itself.

This may be the unifying Cartesian contribution: not information theory in the strict historical sense, but the systematic conversion of an inaccessible condition into a manipulable inscription. The coordinate tells another person where an unseen object is; the cogito attempts to locate an otherwise uncertain subject within an epistemic space. Both achieve certainty by choosing a frame, and both risk disguising the chosen frame as a property of reality.

The strongest formulation would consequently be:

Descartes’s decisive contribution was coordinatization: the construction of symbolic frames in which relations could be encoded, transmitted, manipulated, and reconstructed.

That accommodates analytic geometry, the method of doubt, the cogito, and the warning about primitives. His enduring invention may not have been a particular answer about mind or space, but a method for making a domain calculable by first deciding what will count as an address within it.

REFERENCES

- [1] Galen Strawson, “cogito, ergo sum,” *Galen’s Substack*, 17 December 2025. <https://gstrawson.substack.com/p/cogito-ergo-sum>